Cell Biology Homework Assignment 1

Name _____ Student ID_____

Part 1. True or False (5 points per question, 20 points).

1. Endocytosis is a process involving fusion of secretary vesicles or recycling vesicles with the plasma membrane. **False**

2. Actin is a component of cytoskeletal elements. **True**

3. Plectin connects microfilaments with Intermediate filaments. **True**

4. The smooth ER lumen is the site of synthesis of nearly all of the cell's major classes of lipids. **False**

5. Most proteins synthesized in the rough ER are glycosylated by the addition of a common N-linked oligosaccharide to aspartic acid. **False**

Part 2. Essay Questions (20 points per question, 80 points).

Question 1. Diagram and correctly label the organization of the nucleolus and how it is region is involved in the generation of a ribosome.

The nucleolus contains three regions the Nucleolus organizer retention (NOR), which contains the chromosome regions that harbor the ribosomes. (5 point)

The Fibrillar Center in which rRNAs are transcribed from the genome. (5 point)

The Dense fibrillar component in which RNAs start to be assembled with ribosomal proteins to form immature ribosome subunits. (5 point)

The Granular component, which contains the newly transcribed rRNA and ribosomal proteins. (5 point)

20 points

Question 2. The drug Taxol is just as pernicious to dividing cells as colchicine, and both are used as anti-cancer drugs. Firstly, Taxol binds tightly to microtubules and stabilizes them. When added to cells, Taxol promotes much of the free tubulin to assemble into microtubules. However, the drug colchicine prevents microtubule formation. Based on your knowledge of microtubule dynamics, please explain why both drugs are toxic to dividing cells despite their opposite actions.

Cell division depends on the ability of microtubules both to polymerize and to depolymerize (5 point).
This is most obvious when one considers that the formation of the mitotic spindle requires the prior depolymerization of other cellular microtubules to free up the tubulin required to build the spindle (5 point).
This rearrangement is not possible in Taxol-treated cells, whereas in colchicine-treated cells, division is blocked because a spindle cannot be assembled (5 point).
On a more subtle but no less important level, both drugs block the dynamic instability of microtubules and would therefore interfere with the workings of the mitotic spindle, even if one could be properly assembled (5 point).

20 points

Question 3. Diagram and correctly label the complete process of protein translation from an mRNA transcript.

Initiation phase: (0.5 points)

Assembly of small ribosome subunit with the initiator tRNA binding to mRNA (0.5 points).

Small subunit with initiator tRNA moves along mRNA until locating an AUG codon in the mRNA sequence (0.5 point).

Addition of the Large ribosomal subunit to the small subunit-mRNA-tRNA complex so that the initiator tRNA is in the P-site of the large ribosomal subunit. (1 Point).

Elongation phase (0.5 points)

Addition of tRNA in the A site of the large subunit of the ribosome based on the codon sequence of the mRNA. (1 point)

Formation of a peptide bond between the amino acids at the end of the tRNAs. (0.5 point)

Release of the amino acid from the tRNA in the P site of the large subunit. (0.5 points)

Large subunit shifts one codon in the mRNA so that the current tRNAs shift to the E and P sites. (0.5 points)

Shifting of the small ribosomal subunit to eject the tRNA from the E site from the ribosome. (0.5 points)

Addition of the next tRNA in the free A site and the formation of another peptide bond between the next amino acid at the end of the tRNA. (1 point).

The process continues until the ribosome reaches a specific codon with a stop sequence. (0.5 points).

Termination phase (0.5)

Binding of a release factor on the stop codon so that no new peptide bond forms. (0.5 points)

Instead, a water molecule is added to end the translation process and cause disassembly of ribosome from the mRNA. (0.5 point).

20 points

Question 4. Diagram and correctly label the complete action potential of a neuron. Include the different open and closed states of the associated voltage-gated Na+ channel.

Resting membrane potential (0.5 points)

Activation of a Na+ channel to allow Na+ to enter the cell along its electrochemical gradient (1 point).

Depolarizing phase (0.5 points)

If enough channels are activated, then this will overcome a potential threshold and trigger an action potential (1 point).

Repolarizing phase (0.5 points)

Na+ channels become refractory by closing in a manner that is independent from the voltage stimulus (1 point).

K+ channels open to allow K+ to leave the cell along its concentration gradient (1 points).

Hyperpolarizing phase (0.5 points)

The loss of intracellular K^+ reverses the voltage changes, and the membrane potential is restored to is resting membrane potential fate. (1 point).

Repolarizing phase (0.5 points)

Na+/K+ exchangers remove intracellular Na^+ in exchange for K^+ to restore original resting membrane potential (1 point).

Voltage-gated Na+ restored to original “resting” state (0.5 points).

20 points